KL UNIVERSITY • BACHUPALLY CAMPUS, HYDERABAD

Department of Computer Science & Engineering

# Plant Leaf Disease Detection Using Machine Learning

*Keep it under 12 words - state the problem, not the tool*

PRESENTED BY

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UNDER THE GUIDANCE OF

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AGENDA

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SECTION 1

# Introduction & Domain Background

### What is the domain?

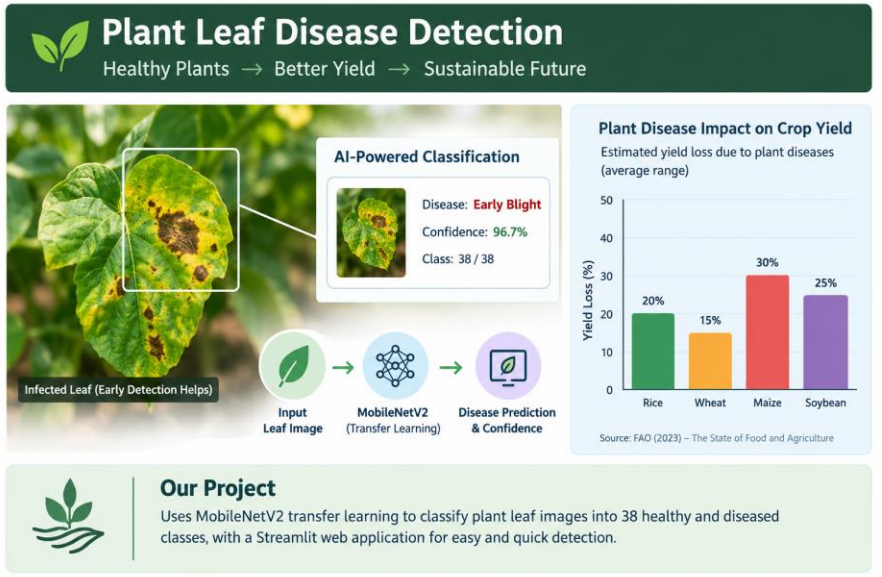
Agriculture is the domain of this project, specifically plant health monitoring and disease detection. Deep learning and image processing are used to identify visual symptoms of plant diseases from leaf images.

### Why is it relevant today?

Plant diseases can reduce crop yield, quality, and farmers' income. Automated image classification can support faster preliminary identification and help farmers decide when to seek expert advice. The project evaluates its model on 10,861 test images from the PlantVillage dataset. (Project evaluation results; dataset source: PlantVillage.)

### Where does your project sit?

The project uses MobileNetV2 transfer learning to classify plant leaf images into 38 healthy and diseased classes, displaying predictions and confidence scores through a Streamlit web application.




**Panel tip**

Every claim on this slide must be traceable to a reference on the last slide.

SECTION 2

# Problem Statement & Motivation

PROBLEM STATEMENT

Manual identification of plant leaf diseases can be difficult and time-consuming for non-experts, creating a need for an automated image-classification system that provides preliminary disease predictions quickly and conveniently.

1

### Who is affected?

- Farmers and agricultural workers
- Students and researchers
- Users who need preliminary plant disease identification

2

### What is the cost of the problem?

- Delayed identification of possible plant diseases
- Potential crop-quality and productivity losses
- Difficulty identifying diseases without expert guidance

3

### Why solve it now?

Machine learning enables automated image classification. Our project uses MobileNetV2 transfer learning to classify leaf images into 38 classes and provides predictions through a Streamlit web application.

SECTION 3

# Literature Survey

*Minimum 8-10 papers studied; show 4-5 of the most relevant here (2020 onwards, journals/conferences only - no blogs).*

| S.No | Author & Year | Title / Technique Used | Dataset / Tools | Key Findings | Limitation / Gap |
|---|---|---|---|---|---|
| 1 | Sharma et al. (2024) | Leaf disease detection using a CNN (ResNet-50) | PlantVillage (54k images) TensorFlow | 96.4% accuracy on 14 crop classes | Fails on field images with mixed lighting |
| 2 | Mohanty et al., 2016 | Using Machine Learning for Image-Based Plant Disease Detection | PlantVillage; deep CNNs | Demonstrated deep learning for plant disease classification using over 50,000 images. | Controlled dataset images may not represent real field conditions. |
| 3 | Gui et al., 2021 | Towards Automatic Field Plant Disease Recognition | Field plant images; CNN | Studied plant disease recognition under field conditions and proposed improvements for field-image classification. | Field images contain complex backgrounds and variable lighting. |
| 4 | Sen & Prerna, 2026 | MobileNetV2-Based Transfer Learning for Real-Time Plant Disease Detection | PlantVillage; MobileNetV2 | Reports 94.2% validation accuracy and 93.6% macro F1-score. | Field-image generalization and robustness remain challenges. |

***The last column is the most important one - your research gap is built from it.***

SECTION 4

# Research Gap Identified

GAP 1

Most plant leaf disease detection models are trained on controlled datasets such as PlantVillage. Their performance may decrease when tested on real farm images with varying lighting, complex backgrounds, and different leaf conditions.

GAP 2

Many plant disease detection approaches focus on model development and accuracy, with limited emphasis on providing a simple interface for users to upload leaf images and obtain predictions easily.

GAP 3

Many plant disease detection models provide a predicted disease label without clearly explaining the prediction or ensuring reliable performance on unseen, real-world images.

OUR PROJECT ADDRESSES

Our project primarily addresses GAP 2 by providing a user-friendly Streamlit web application that predicts plant leaf diseases and displays the predicted class with a confidence score.

*A gap is a sentence that begins with "No existing work has..." or "Existing work does not..."*

Rule: every gap here must come from the last column of the literature survey table. A gap you cannot trace to a paper will be rejected by the panel.

SECTION 5

# Objectives of the Project

*Write 3-4 objectives. Each one must start with an action verb and be something you can demonstrate by the final review.*

1. To collect and prepare the PlantVillage dataset, containing approximately 1.3 GB of leaf images across 38 healthy and diseased classes, sourced from the PlantVillage dataset repository on GitHub.

2. To design and implement a MobileNetV2-based transfer learning model that classifies plant leaf images into 38 healthy and diseased classes, addressing automated disease prediction through a user-friendly Streamlit application.

3. To evaluate the model using accuracy, precision, recall, and F1-score; achieved 95.35% test accuracy on 10,861 test images across 38 classes.

4. To deploy and demonstrate a working Streamlit web application that allows users to upload plant leaf images and view the predicted disease class along with its confidence score.

*Tip: objectives and results must match one-to-one at the final review.*

SECTION 6

# Proposed System Architecture

**INPUT**

Users upload a plant leaf image through the Streamlit web application to receive a disease prediction

**PRE-PROCESSING**

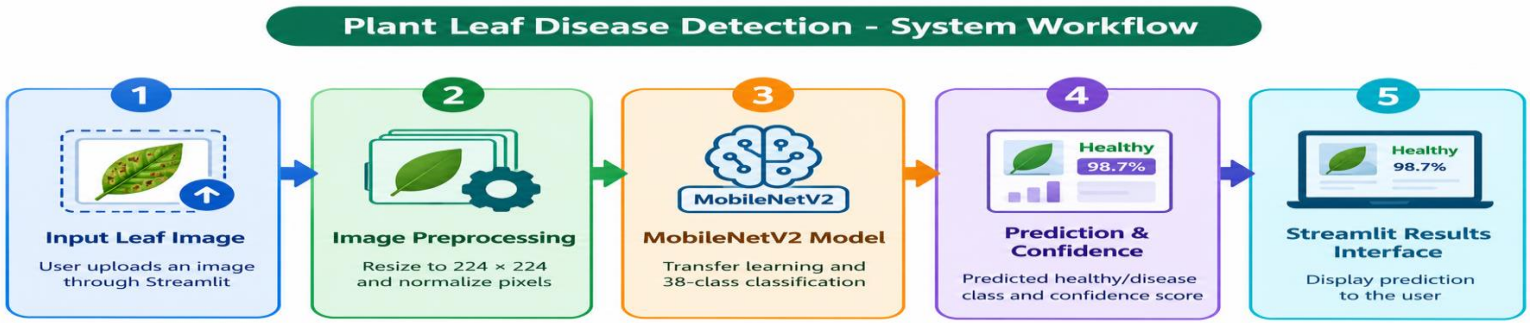
Clean the dataset, resize leaf images to 224 × 224 pixels, normalize pixel values, encode class labels, and split the data into training, validation, and testing sets.

**CORE MODULE**

MobileNetV2-based transfer learning using TensorFlow/Keras to classify plant leaf images into 38 healthy and diseased classes.

**OUTPUT**

Display the predicted plant disease class and confidence score in the Streamlit UI, with model evaluation results of 95.35% accuracy, 0.95 macro precision, 0.94 macro recall, and 0.94 macro F1-score.




**Label every arrow**

Each arrow should say what flows through it - raw image, cleaned CSV, feature vector, JSON response. Unlabelled arrows invite questions you cannot answer.

SECTION 7

# Methodology & Modules

## 1 Module 1 - Data

PlantVillage dataset (~1.3 GB), downloaded from the public GitHub repository. It contains leaf images covering 38 healthy and diseased classes. Images were preprocessed and split into training, validation, and testing sets; the test set contains 10,861 images.

## 2 Module 2 - Processing

Cleaned the dataset, resized images to 224 × 224 pixels, normalized pixel values, and encoded class labels. MobileNetV2 extracts image features for classification. Data augmentation may be used during training to improve generalization, if implemented in the project.

## 3 Module 3 - Core Logic

Algorithm: MobileNetV2-based Transfer Learning was chosen because it is lightweight, computationally efficient, and suitable for image classification. Compared with training a CNN from scratch, it reuses pretrained features, reducing training time while achieving 95.35% test accuracy on the PlantVillage dataset.

## 4 Module 4 - Interface

**Frontend:** Streamlit web interface for uploading leaf images and displaying predictions.
**Backend:** Python, TensorFlow/Keras, and the trained MobileNetV2 model for image preprocessing and disease classification.
**How to Run:** Activate the Python virtual environment and execute python -m streamlit run src/app.py in the terminal. Open the displayed local URL, upload a leaf image, and view the predicted class and confidence score.

SECTION 8

# Tools and Technologies

## SOFTWARE

- **Language:** Python 3.11
- **Libraries:** TensorFlow, Keras, NumPy, scikit-learn, Pillow
- **Front end:** Streamlit
- **Back end:** Python with TensorFlow/Keras for image classification
- **Database:** Not required
- **IDE:** Visual Studio Code (VS Code)

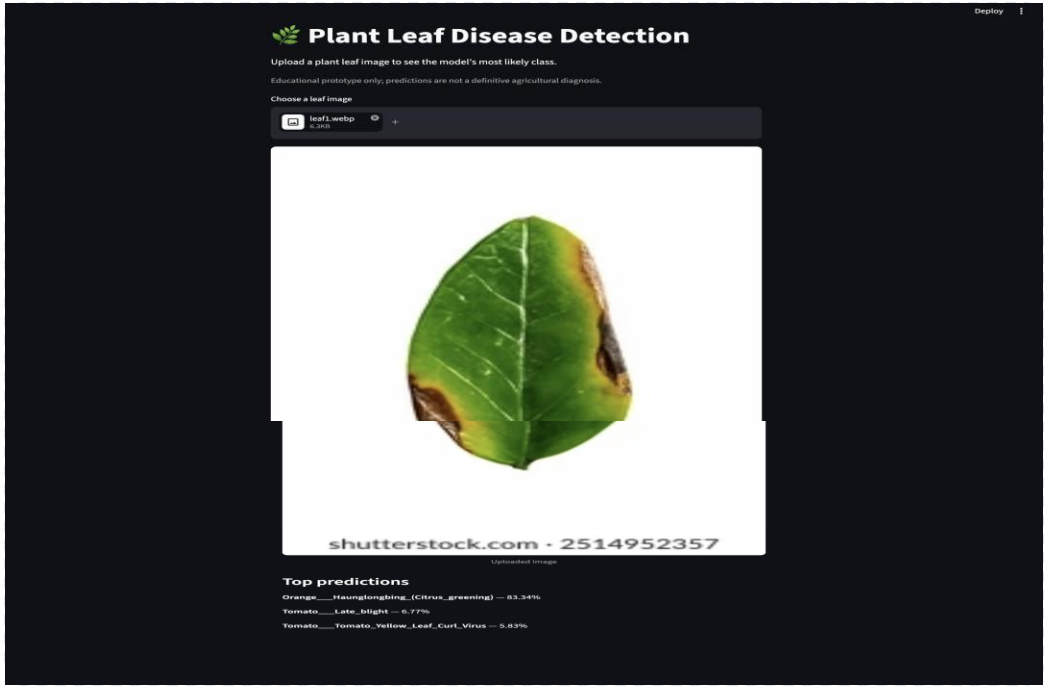
## HARDWARE / PLATFORM

- **Processor:** Intel Core i5 or equivalent and above
- **RAM:** Minimum 8 GB
- **GPU / Cloud:** CPU-based local execution or Google Colab GPU
- **Sensors or Devices:** None required; user uploads leaf images
- **Deployment:** Localhost using Streamlit
- **Version Control:** GitHub

SECTION 9

# Work Completed / Demo

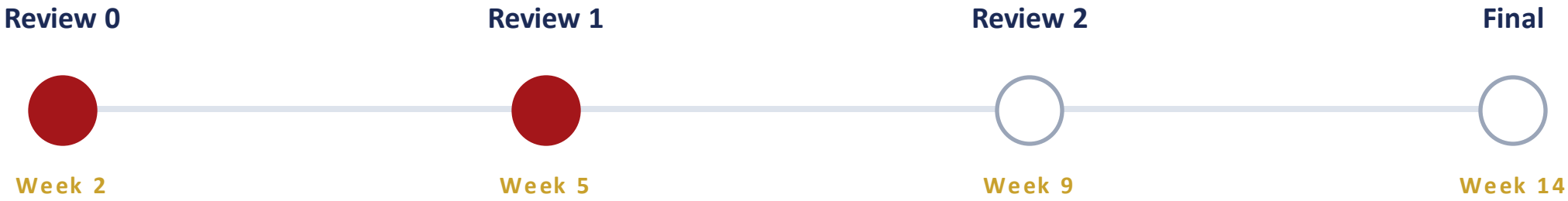
| | |
|---|---|
| Literature survey | Completed |
| Dataset collection | Completed |
| Pre-processing pipeline | Completed |
| Model training | Completed |
| User interface | Completed |
| Testing & evaluation | Completed |




Show one working thing, however small. A live screenshot beats three slides of claims.

SECTION 10

# Timeline & Review Plan




| Review 0 | Review 1 | Review 2 | Final |
|---|---|---|---|
| Week 2 | Week 5 | Week 9 | Week 14 |
| Title, abstract, team roles finalised | Literature survey, gap, objectives | Architecture, 60% implementation | Full demo, results, report, paper draft |

*Filled circles = completed reviews. Update this slide before every review.*

**What the panel checks every time:** progress since the last review, honesty of the status you claim, and whether each objective still has enough weeks left to finish.

SECTION 11

# Expected Outcome & Conclusion

### DELIVERABLE

A working system that detects and classifies plant leaf diseases from uploaded images using MobileNetV2 deep learning and displays predictions with confidence scores through a Streamlit web application.

### MEASURABLE TARGET

Target: Achieve test accuracy above 90%; the current model has achieved 85.35% test accuracy on 10,861 images across 38 classes.

### BEYOND THE LAB

**Target Users:** Farmers, agricultural students, researchers, and agricultural extension workers.
**Deployment:** Localhost using Streamlit, with potential future deployment on a cloud platform for wider accessibility.

### CONCLUSION

Plant diseases can reduce crop health and productivity, making early identification important for farmers and agricultural users. The literature gap is that models trained on controlled datasets may not generalize well to real-world field images, and users need a simple interface for disease prediction. Our system delivers a MobileNetV2-based Streamlit application that classifies 38 plant leaf classes and achieved 95.35% test accuracy on 10,861 test images.
**Future scope:** Real-time disease detection from live camera images and validation under diverse field conditions are left out of the current scope.

SECTION 12

# References

*IEEE style, numbered in the order they are cited in the slides. Minimum 8 references for a minor project - these are format examples only.*


**1.** Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using Deep Learning for Image-Based Plant Disease Detection. Frontiers in Plant Science, 7, 1419. https://doi.org/10.3389/fpls.2016.01419
DOI
**2.** Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 4510–4520. https://arxiv.org/abs/1801.04381
arXiv
+1
**3.** PlantVillage Dataset. Plant leaf disease image dataset and corresponding labels. GitHub repository. https://github.com/spMohanty/PlantVillage-Dataset
GitHub
**4.** TensorFlow. MobileNetV2 documentation. https://www.tensorflow.org/api_docs/python/tf/keras/applications/MobileNetV2
**5.** Streamlit. Streamlit documentation. https://docs.streamlit.io/


***Every reference listed here must appear in the literature survey table, and every row of that table must appear here.***

# Thank You

Questions & Suggestions

Team: Plant Leaf Disease Detection | Guide: Dr. G. S. Raghavendra | Dept. of CSE, KLH Bachupally